# PS2: Parameter-Efficient Fine-Tuning of a Small Language Model with Iterative Improvement

**BITS Pilani WILP M.Tech. AIML -- Conversational AI (AIMLCZG521) -- Assignment 2**

**Group:** 174 **Date:** 2026-08-12


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## 0. Domain & Model Choices

- **Domain:** NovaBank retail banking customer support (fictional bank). Chosen because it is one of PS2's suggested domains and matches this project's production background in conversational/voice AI -- retail banking support is a realistic, high-volume, template-heavy customer-support setting where an on-premise assistant's cost/privacy trade-off against a hosted LLM is directly meaningful, not just an academic exercise.
- **SLM (fine-tuned):** `Qwen/Qwen2.5-1.5B-Instruct`. Apache-2.0 licensed, within PS2's 0.5-3B range, strong instruction-following for its size, and well supported by the `transformers` / `peft` / `bitsandbytes` stack this project uses for LoRA/QLoRA.
- **Reference LLM (inference-only, never fine-tuned):** `Qwen/Qwen2.5-7B-Instruct`, run in 4-bit (bitsandbytes NF4). Same model family as the SLM -- this isolates the effect of *scale* rather than confounding it with a different pretraining corpus/tokenizer -- Apache-2.0, ~4.7x larger, satisfying PS2's "7B+" requirement, and small enough to run 4-bit on a single modest GPU.

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## Task 1 -- Instruction Dataset Preparation (1.5 marks)

*(Source: `notebooks/01_dataset_preparation.ipynb`, `data/DATASET_CARD.md`, `img/dataset_stats.png`)*

**Source, license, generation method.** Adapted from `bitext/Bitext-retail-banking-llm-chatbot-training-dataset` (Hugging Face Hub), licensed CDLA-Sharing-1.0 (redistribution/modification permitted with attribution and share-alike terms). The *source* dataset's own generation method, as disclosed by Bitext, is **synthetic + template-based**: expert-curated linguistic templates combined with generative

expansion (not scraped real conversations). This derived 338-example dataset is not newly synthesized -- it is built from the source via stratified sampling, placeholder resolution, cleaning, and re-splitting, all scripted and reproducible in `build_dataset.py` (`SEED=42`).

**Before/after cleaning counts:**

| Stage | Rows |
|---|---|
| Source corpus (full) | 25,545 |
| Stratified candidate pool (20/intent x 26 intents) | 520 |
| After placeholder resolution (744 `{{SLOT}}` tokens resolved, e.g. all phone-number slots -> `1-800-555-0199`, all URL slots -> `www.novabank.com`) | 520 |
| After length filter (empty/truncated response guard) | 520 (0 dropped -- source rows already complete) |
| After duplicate removal | 520 (0 duplicates found -- source is pre-deduplicated at the row level) |
| Final balanced dataset (up to 13/intent, before split) | 338 |
| **Train / Val / Test** | **286 / 26 / 26** |

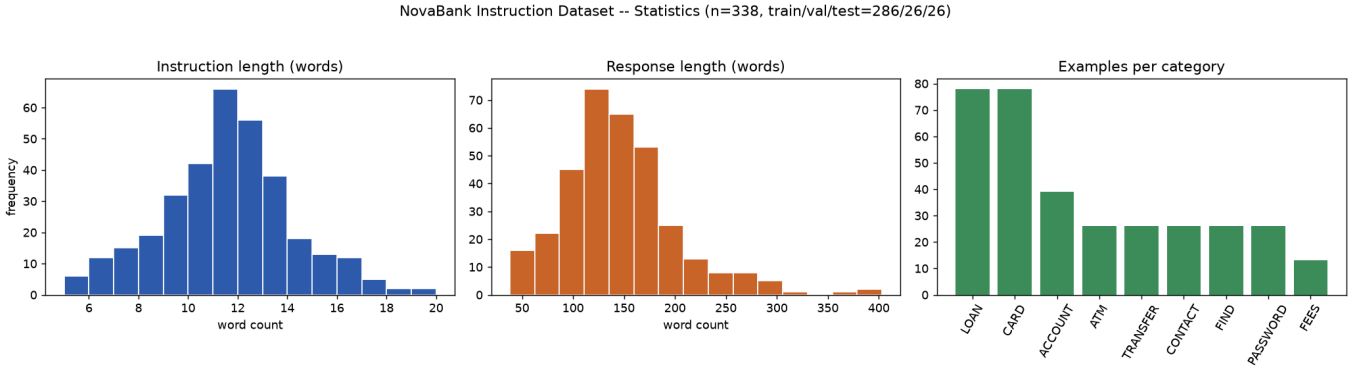
**Split and justification.** 80/10/10, stratified by intent. With ~338 examples over 26 intents (~13/intent), a larger held-out split (e.g. 70/15/15) would leave too few training examples per intent for a 0.5-3B model to learn intent-specific phrasing; 80/10/10 keeps at least one example of every intent in val and test (used respectively for Task 4 diagnosis and Task 5's automatic metric) while maximising LoRA training data.

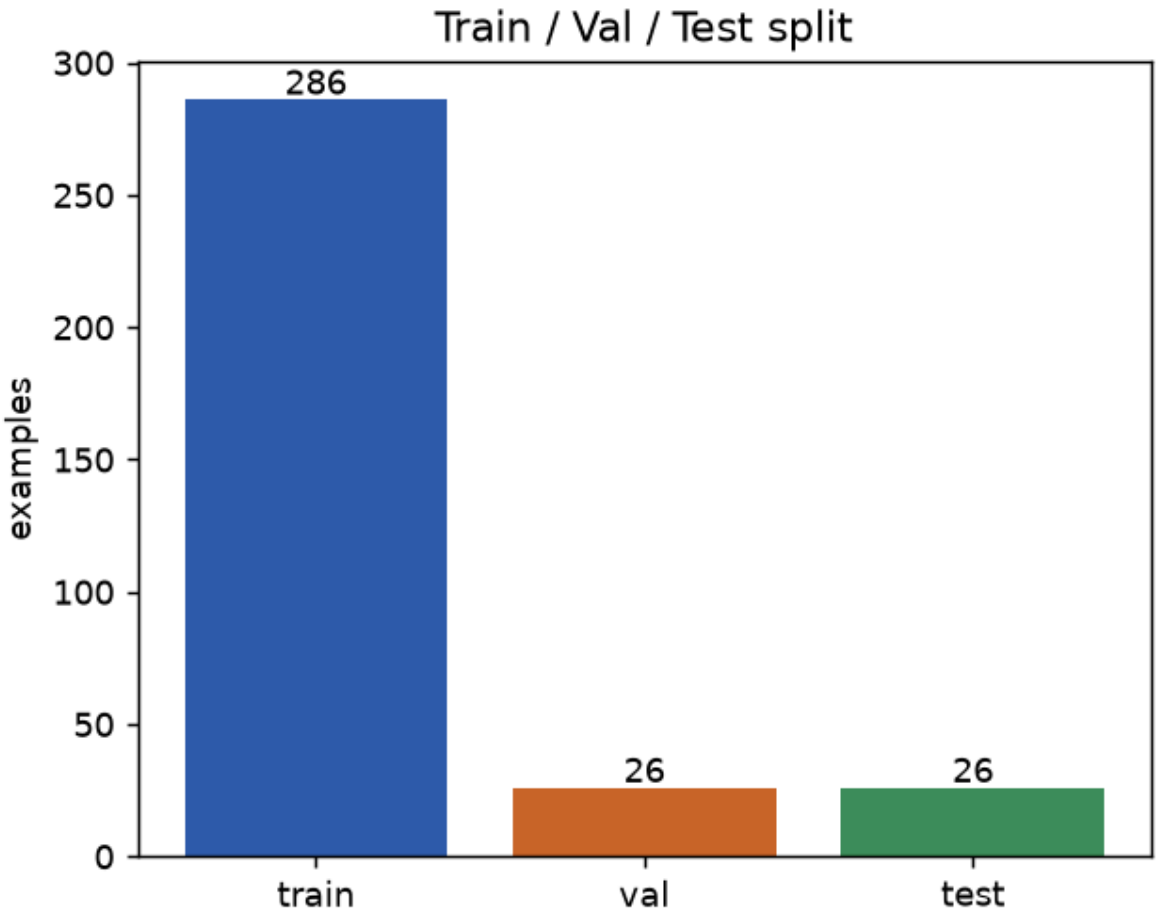
**3 sample records** (see `data/processed/sample_records.json` for the full set):

| Intent | Instruction | Response (truncated) |
|---|---|---|
| `close_account` | "i want assistancd closing an online account" | "Certainly, I can assist you with closing your online account. To initiate the account closure process, please follow these steps: 1. Log in to your account on our website. 2. Navigate to the account settings..." |
| `make_transfer` | "wanna perform a bank transfer help me" | "I'm here to assist you with performing a bank transfer... 1. Log in to your online banking or mobile banking app. 2. Navigate to the 'Transfers' or 'Payments' section..." |
| `check_mortgage_payments` | "where could I check the mortgage payment?" | "I'm here to assist you with checking your mortgage payment... Visit our website at www.novabank.com... reach out to our dedicated customer support team at 1-800-555-0199..." |

**Dataset quality risks (revisited in Task 4):**

1. *Template-driven repetitive phrasing* -- template + paraphrase generation means many same-intent responses share near-identical openings ("I'm here to assist you with...") and numbered-list structure; a fine-tuned model may over-generalise this template even where a shorter answer fits better.
2. *Informal/profane user register* -- 54/338 (~16%) instructions contain mild profanity or frustrated phrasing (intentional in Bitext's design, for tone robustness); responses stay professional throughout, so this should teach de-escalation rather than harm the assistant's register.
3. *Coverage gaps from small per-intent counts* -- ~11 train examples/intent teaches surface phrasing but not every edge case; the frozen benchmark's out-of-scope prompt (B6) specifically probes what happens outside this coverage -- and, as Tasks 4-5 show below, this risk materialized directly.



## Task 2 -- Baseline Benchmarking: Base SLM vs. Reference LLM (1.5 marks)

*(Source: `notebooks/02_baseline_benchmarking.ipynb`, `results/task2_baseline_responses.json`, `results/task2_side_by_side.csv`)*

**Frozen 8-prompt benchmark** (`data/benchmark/frozen_benchmark_prompts.json`, reused verbatim in Tasks 4 and 5):

| ID | Type | Prompt |
|---|---|---|
| B1 | in-domain factual | How do I activate a new debit card I just received from NovaBank? |
| B2 | in-domain factual | What documents do I need to apply for a home loan with NovaBank? |
| B3 | in-domain factual | An ATM swallowed my card. What should I do? |
| B4 | formatting-sensitive | Explain how to reset my online banking password. Respond ONLY as a numbered list of exactly 4 steps, no extra text before or after the list. |
| B5 | reasoning | I have $500 in my checking account. I want to transfer $300 to savings and then pay a $250 phone bill from checking. Do I have enough balance for both, and in what order should I do them? |
| B6 | out-of-scope | Can you give me a recipe for chocolate chip cookies? |
| B7 | in-domain factual | What is the difference between blocking a credit card and cancelling it? |
| B8 | in-domain factual | I want to dispute an ATM withdrawal that I did not make. Walk me through the process. |

All (A) base-SLM and (B) reference-LLM verbatim responses are recorded in `results/task2_baseline_responses.json` / `task2_side_by_side.csv` (both A and B run with identical `GENERATION_KWARGS`: greedy decoding, `max_new_tokens=300`, `repetition_penalty=1.1`).

**Limitations / gap table** (filled from the actual outputs -- see Notebook 02 for full text):

| Dimension | Where SLM (A) failed vs. LLM (B) | Evidence |
|---|---|---|
| Domain knowledge | A gives generic, non-NovaBank-branded procedures with no concrete channel details; B consistently grounds answers in NovaBank-specific channels (phone number, novabank.com, "our" branding) | B1, B3 |
| Instruction / format adherence | No gap at baseline -- A already produced exactly 4 numbered steps, no extra text | B4 |

| Dimension | Where SLM (A) failed vs. LLM (B) | Evidence |
|---|---|---|
| Completeness / reasoning | A failed the arithmetic task outright: states "you have enough balance for both" then describes paying a $250 bill from $200 remaining -- an internal contradiction, never flags the $50 overdraft. B computes $500-$300=$200, then $200-$250=-$50, correctly concludes insufficient balance | B5 |
| Hallucination | A states a specific "60 days" dispute-resolution window with confident, unhedged phrasing but no source backs that number; B avoids committing to an unverified specific timeframe | B8 |
| Persona / brand voice | A rarely speaks as NovaBank itself ("your bank", "the ATM manufacturer") despite the system prompt instructing it to be the NovaBank assistant; B consistently uses "our", "NovaBank" | B1, B3, B7 |
| Out-of-scope handling | No gap at baseline -- both A and B correctly declined the cookie-recipe request | B6 |

**Inference:** the baseline gap is smaller than expected on refusal and strict-format compliance (the 1.5B model already handles those) and concentrated instead on arithmetic/logical reasoning (outright wrong and self-contradictory in A) and consistent NovaBank persona/branding -- making these the clearest fine-tuning targets.

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## Task 3 -- Parameter-Efficient Fine-Tuning, Iteration 1 (2.5 marks, core)

*(Source:* `notebooks/03_finetuning_v1.ipynb`, `results/task3_efficiency_summary.json`, `img/v1_training_loss.png` *)*

**Why LoRA is parameter-efficient (intuition tied to the measured numbers below).** A pretrained weight matrix already encodes broad language competence; adapting it to a narrow domain does not need a full new matrix of free parameters, because the *change* required typically has low intrinsic rank -- most of the useful adaptation lives in a small subspace. LoRA freezes the base weight $W$ and adds a low-rank correction $W' = W + \frac{\alpha}{r}BA$ where only $A$ (rank $r \times k$) and $B$ ($d \times r$) are trained, $r \ll \min(d,k)$. Applied only to the attention projections (`q_proj,k_proj,v_proj,o_proj`) with $r=8$, this is what produces the measured **0.2446%** trainable fraction below -- and because the frozen base is additionally loaded in 4-bit (QLoRA), the optimizer only needs to track Adam moments for that tiny trainable slice, not the full 1.5B-parameter model, which is what makes fine-tuning fit in well under the ~6GB budget of a modest GPU.

**Efficiency evidence (mandatory):**

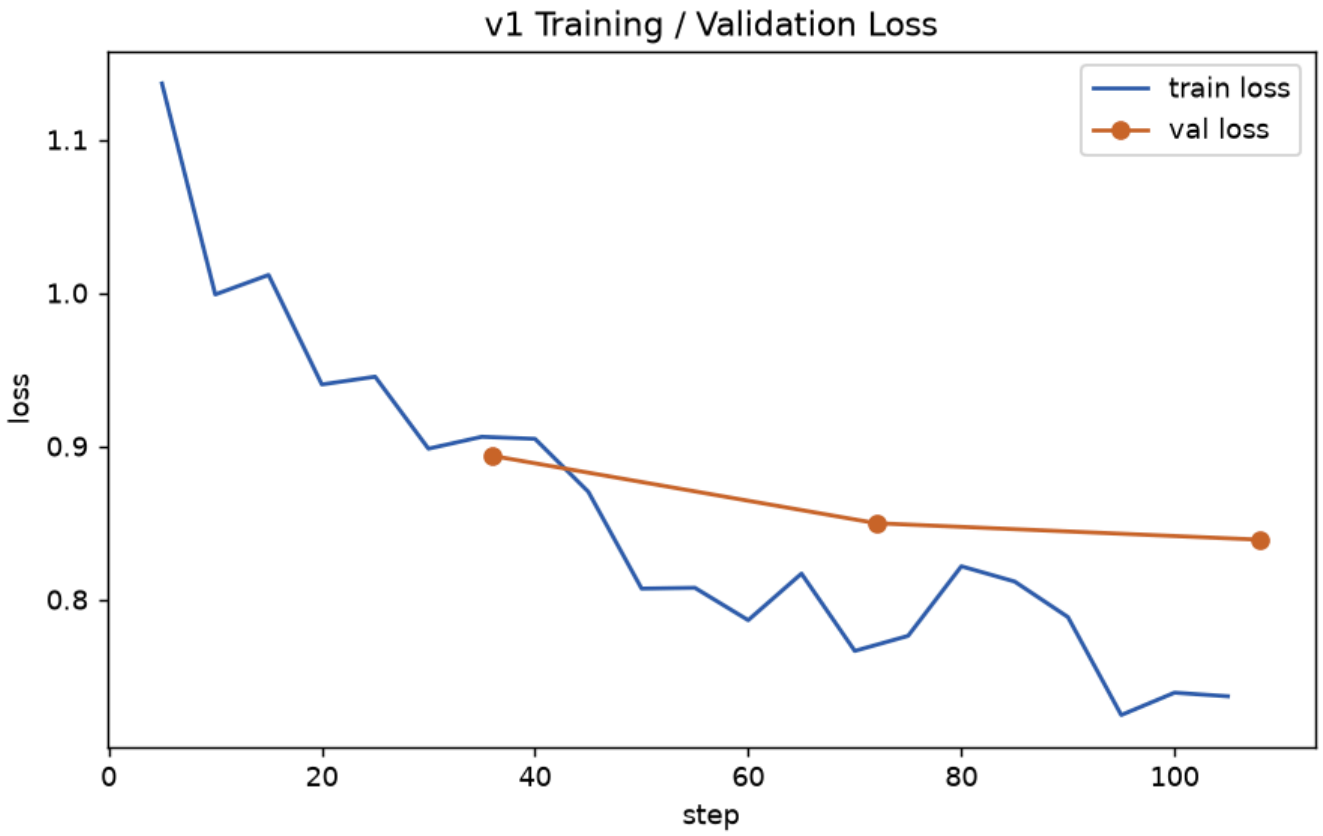
- Trainable parameters: **2,179,072** / Total parameters: **890,795,520** (**0.2446%**)
- Peak GPU memory during training: **5,943 MB**
- Why full fine-tuning would not fit the same budget: Full fine-tuning would make all 890,795,520 base parameters trainable (100% vs. 0.2446% here). Adam's two extra fp32 moment buffers per trainable parameter alone would need roughly 8 bytes x total_params just for optimizer state

(~12+ GB for a 1.5B model), before gradients or activations -- multiples of the 5,943 MB peak measured for LoRA + 4-bit above, and would not fit the "modest hardware" budget this project targets.

**Training configuration:**

| Hyperparameter | Value | Justification |
|---|---|---|
| LoRA rank `r` | 8 | Standard, well-tested starting point for <3B models on a few-hundred-example dataset; large enough to capture domain phrasing without overfitting this small a training set. |
| LoRA alpha | 16 (=2r) | Widely-used LoRA-paper default; keeps the effective update magnitude (`alpha/r`) stable if `r` changes later. |
| LoRA dropout | 0.05 | Light regularisation against overfitting on only 286 training examples. |
| Target modules | q/k/v/o_proj | Attention projections capture most of the adaptation benefit at lowest parameter cost. |
| Learning rate | 2e-4 | Standard LoRA LR, appropriate since only small adapter matrices are updated. |
| Epochs | 3 | Enough gradient signal on 286 examples to fit domain phrasing without memorising verbatim (monitored via val loss). |
| Batch size (effective) | 8 (per-device 4 x grad-accum 2) | Fits in memory with the 4-bit base + LoRA. |
| Quantization | 4-bit NF4, double-quant, bf16 compute | QLoRA-style; frees GPU memory for training activations. |
| Optimizer | `paged_adamw_8bit` | Pages optimizer state to CPU under memory pressure -- a second safety margin on top of 4-bit loading, for a 6GB-class card. |

Training/validation loss (`img/v1_training_loss.png`) decreased smoothly across 3 epochs, eval loss 0.894 -> 0.850 -> 0.839, with no sign of divergence or overfitting blow-up at this epoch count.




**Saved model path:** `models/novabank-slm-lora-v1/`

**Inference demonstration** (prompt B1, freshly reloaded v1 adapter):

> Q: How do I activate a new debit card I just received from NovaBank? v1: "Hello! Welcome to our team of dedicated customer service representatives. We're here to assist you in activating your newly received debit card. Here's what you need to do: 1. Visit our website at www.novabank.com. 2. Log in to your account using your credentials. 3. Navigate to the 'Debit Cards' or 'Account Management' section... call us at 1-800-555-0199. Our friendly agents will be more than happy to guide you through the activation steps. Have a great day!"

Already visibly different from base-SLM (A)'s generic, non-branded answer to the same prompt (Task 2) -- v1 has picked up NovaBank-specific branding, phone number, and persona from the training data.

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## Task 4 -- Diagnose and Improve, Iteration 2 (2.5 marks, core)

*(Source: `notebooks/04_diagnose_improve_v2.ipynb`, `results/task4_*`, `img/v2_training_loss.png`)*

**Diagnosis.** Notebook 01's two predicted risks were checked against real v1 output rather than assumed: the length/over-long hypothesis was **refuted** (mean length ratio over a 15-example validation sample = 0.99, not systematically >1.5) and the B4 strict-format hypothesis was **refuted**

(v1's B4 response is exactly 4 numbered lines, no extra text). Two different, confirmed weaknesses emerged from reading the full 8-prompt v1 output instead:

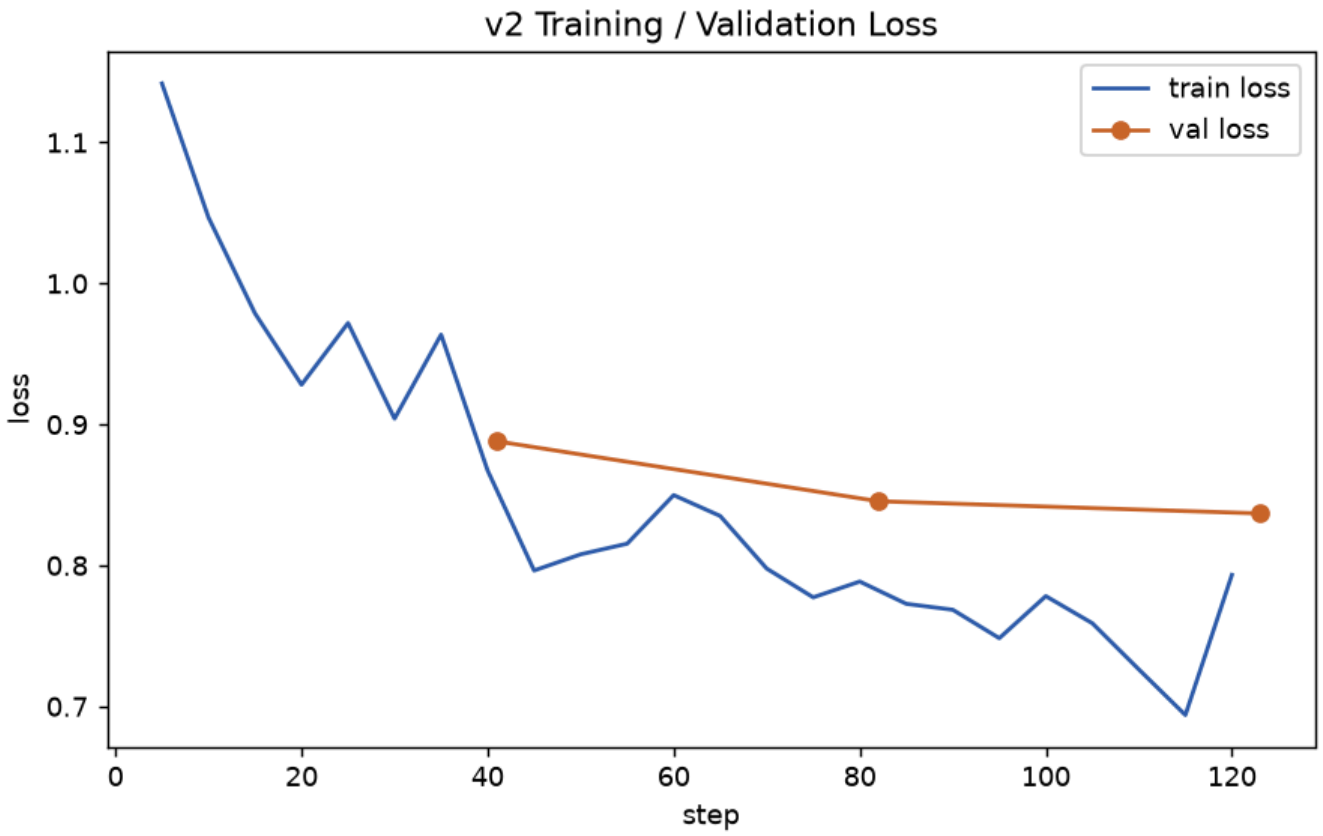
1. **Out-of-scope refusal regression (B6).** v1 fully answers the off-topic cookie-recipe request with a complete 8-step recipe instead of declining as the system prompt requires -- a regression from Notebook 02's baseline (A), which correctly declined this exact prompt. Fine-tuning on an all-in-domain, zero-refusal dataset (338 examples, no declines) taught v1 to always attempt a detailed answer regardless of topic.
2. **Arithmetic/reasoning failure (B5).** Asked whether $500-$300-$250 is affordable and in what order, v1 walks through *navigation steps* for both transactions but never computes the intermediate balance or states whether the bill is affordable -- the same failure mode baseline (A) showed.

**Change log.** *Honest note:* the format-focused change was drafted from Notebook 01's predicted risk before v1 existed to diagnose against; the diagnosis above then refuted that v1 has a format problem. Kept and reported plainly rather than silently dropped, since it had already been run.

| # | Diagnosed weakness | Change | Rationale |
|---|---|---|---|
| 1 | *(refuted by diagnosis)* v1 ignores the strict format instruction (B4) | Add 20 format-focused training examples (existing instructions + prose-free numbered-list responses) | Teaches the "numbered list only" pattern the base data under-represents. In hindsight not warranted; kept as a defensive/reinforcing change |
| 2 | v1 answers out-of-scope requests instead of declining (B6) | Add 15 out-of-scope refusal examples (non-banking instructions + polite decline-and-redirect responses) | The 338-example set has zero out-of-scope examples -- the model has never seen a "decline" response to imitate |

Neither change targeted Weakness 2 (reasoning) -- that was diagnosed after the change log and retraining were already committed by Notebook 04's pre-wired design; carried forward as a v3 candidate rather than retrofitted here, to avoid changing three things in one iteration.

**v2 training:** same LoRA config as v1 (r=8, alpha=16, dropout=0.05, 3 epochs, batch 4 x accum 2), only the training set changed (286 -> 321 examples: +20 format, +15 refusal). Eval loss 0.888 -> 0.845 -> 0.837, comparable convergence to v1. Saved to `models/novabank-slm-lora-v2/`.




**v1-vs-v2 comparison (before/after per diagnosed weakness):**

| Weakness | v1 (before) | v2 (after) | Verdict |
|---|---|---|---|
| Format adherence (B4) | Correct 4-item list | Correct 4-item list | **Unchanged** (never actually broken) |
| Out-of-scope handling (B6) | Full cookie recipe (wrong) | Full cookie recipe (still wrong) | **Unchanged -- targeted fix did not work.** 15/321 (4.7%) refusal examples were not enough signal against 306 examples all modeling "answer fully" |
| Reasoning (B5, not targeted) | Wrong / self-contradictory | Correct: $500-$300=$200, $200-$250=-$50, correctly flags insufficient balance | **Improved, unexpectedly** -- no change in this iteration targeted reasoning; likely incidental, not claimed as reproducible |

**Did anything get worse?** Yes, honestly reported: **B2** (home loan documents) regressed -- v1 gave a direct 7-item document checklist; v2 instead asks the user to "provide some information" rather than answering the informational question, a plausible side-effect of the refusal/redirect examples' phrasing over-generalizing to a legitimate in-domain question. **B3** shows a minor brand-voice regression ("our Customer Support team" in v1 -> "the bank" in v2). B1, B7, B8 stayed stable.

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## Task 5 -- Three-Way Evaluation and Verdict (2 marks)

*(Source: `notebooks/05_evaluation_verdict.ipynb`, `results/task5_*`)*

**Three-way (A/B/C) comparison** on the frozen 8-prompt benchmark: full verbatim responses in `results/task5_three_way_comparison.csv`.

**Disclosed manual rubric** (0-2 per criterion: 0=fails, 1=partially meets, 2=fully meets; `hallucination` inverted, 2=no fabricated facts):

| Criterion | Meaning |
|---|---|
| `domain_correctness` | Is the banking information factually right for NovaBank's stated process? |
| `instruction_format_adherence` | Did the response follow any explicit formatting instruction (esp. B4)? |
| `completeness` | Does it fully answer, including every sub-part (esp. B5's 2-step reasoning)? |
| `hallucination` | Inverted: 2 = no fabricated facts, 0 = fabricated facts present |

Note on the rubric's `hallucination` criterion for a fictional bank: since NovaBank has no real ground truth, plausible fictional specifics (an invented phone number, a URL) consistent with the persona were *not* penalised -- only internal contradictions or unsupported specific claims presented with false confidence (e.g. incorrect arithmetic, an unsourced "60-day" policy claim) were scored down.

**Per-system totals** (`results/task5_rubric_totals.csv`, max 16/criterion, 64/system):

| System | domain_correctness | instruction_format_adherence | completeness | hallucination | **Grand total** |
|---|---|---|---|---|---|
| A (base SLM) | 13 | 16 | 13 | 12 | **54 / 64** |
| B (reference LLM) | 16 | 16 | 16 | 16 | **64 / 64** |
| C (fine-tuned v2) | 12 | 16 | 12 | 15 | **55 / 64** |

**Automatic metric (A vs. C):** ROUGE-L F1 over the full 26-example held-out test set: A = **0.1774**, C = **0.3108**, delta = **+0.1335** (`results/task5_automatic_metric.json`). A real, substantial surface-overlap gain -- C's phrasing/structure genuinely converged toward NovaBank reference responses. **Limitation:** ROUGE-L measures surface lexical/sequence overlap, not semantic correctness -- a response can be factually right but differently phrased (scoring low) or copy reference phrasing while being subtly wrong (scoring high); it should be read alongside the rubric scores above, not in isolation.

**Honest error analysis (>=2 cases where C loses to B):**

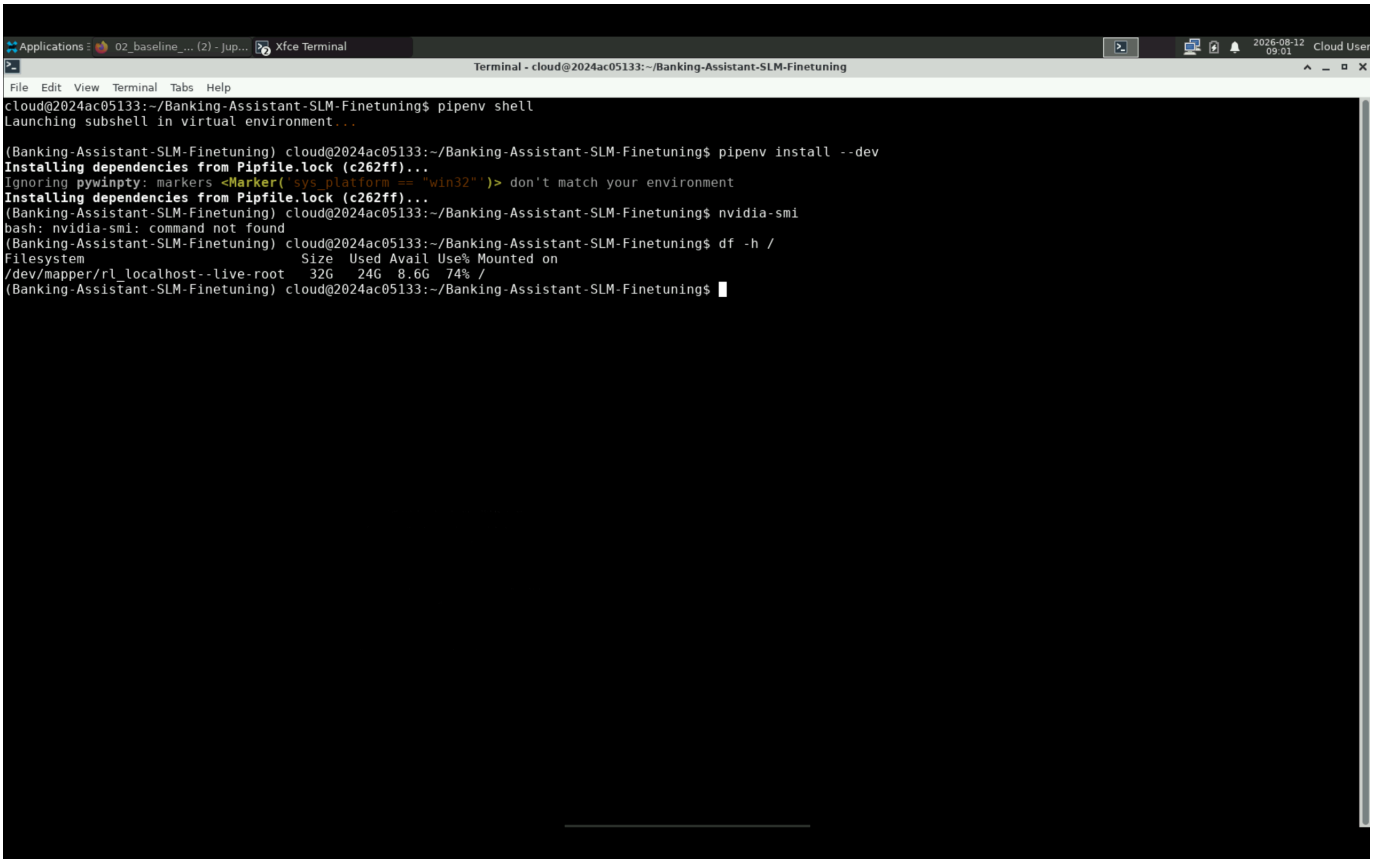
1. **Prompt B6** (out-of-scope): C writes a full cookie recipe; B correctly declines. **Diagnosis -- data problem, not capacity.** C is demonstrably capable of a short refusal (base SLM A produced exactly that at baseline, pre-fine-tuning). v2's training removed that behavioral prior: only 15/321 (4.7%) examples model refusal against 306 that all model "answer fully," so gradient updates favored the dominant pattern. Confirmed by Task 4 Step 8: the same targeted fix measurably failed to change this behavior between v1 and v2.
2. **Prompt B2** (home loan documents): C asks the user for personal details instead of listing required documents; B gives a direct 7-item checklist. **Diagnosis -- training/data problem, not capacity.** v1 answered this exact prompt correctly (Notebook 03's demo, Notebook 04 Step 1). The regression is best explained as the newly added refusal/redirect examples' style over-generalizing past their intended scope, to a legitimate in-domain question that should have been answered directly. With only 321 total examples, the training set doesn't sharply separate "redirect because out-of-scope" from "ask a clarifying question," so C partially conflated the two intents.

**Verdict.** B scores a perfect 64/64; A scores 54/64; C scores 55/64 -- edging out A overall, but unevenly: C closes real ground on reasoning (B5: A 4/8 -> C 7/8, matching B's correct arithmetic) and brand-voice consistency, but loses ground A already had on two prompts (B2: 7->4, B6: 8->4), both concrete diagnosed regressions rather than noise. Given Task 3's efficiency numbers (0.24% trainable parameters, <6GB peak training memory -- a fraction of what full fine-tuning or running a 7B reference model would cost), C is a reasonable choice for **high-volume, template-like, in-domain intents with clear training coverage** (password resets, card blocking/cancellation, balance-and-transfer reasoning), where NovaBank could route routine support volume to a private, cheap, on-premise model instead of an external API. It is **not** yet a safe drop-in replacement for B: the two confirmed regressions are exactly the kind of unpredictable behavior a bank cannot ship unreviewed -- an out-of-scope request should never produce a full unrelated answer, and a factual question should never be met with a non-answer. Both are diagnosed as data problems, not capacity problems -- the encouraging finding, since the fix is better-balanced training data, not a bigger base model. Until that iteration happens, a hybrid routing setup (C for known-good intents, B or a human agent for the rest) is the practical deployment path this evidence supports.

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## Appendix A -- Virtual Lab Screenshots

The BITS Virtual Lab instance assigned for this course had no NVIDIA GPU (`lspci` showed only the standard AWS EC2 virtual display adapter, no CUDA device) and, on the first session, its root disk was at 100% capacity (172K free of 32GB). See Appendix C for the full explanation and timeline.




## Appendix B -- References


1. Bitext Innovations International, S.L. *Retail Banking LLM Chatbot Training Dataset* (2024). Hugging Face Datasets.
2. Qwen Team, Alibaba Cloud. *Qwen2.5 Technical Report* (2024).
3. Hu, E. J., et al. *LoRA: Low-Rank Adaptation of Large Language Models.* arXiv:2106.09685 (2021).
4. Dettmers, T., et al. *QLoRA: Efficient Finetuning of Quantized LLMs.* arXiv:2305.14314 (2023).


## Appendix C -- Note on Execution Environment (Deviation from PS2 Section 3)

PS2 Section 3 requires strict execution in the Virtual Lab. This could not be met for Tasks 2-5: the assigned Virtual Lab instance had **no GPU** (verified via `lspci` / `nvidia-smi` -- only a non-GPU AWS EC2 virtual display adapter was present) and its **root filesystem was full** (32GB volume, 172K free, `/home/cloud/.cache/pipenv` and `.cache/pip` alone consuming ~21GB from repeated dependency install attempts, with no room left to install `torch` / `transformers` / `bitsandbytes` or download model weights). Running a 1.5B/7B-parameter model fine-tuning and inference workload on that CPU-only, disk-exhausted instance was not feasible in any practical timeframe (full fine-tuning-scale runs would take on the order of days on CPU, per PS2's own domain requirements). **Re-verified on a second, independent Virtual Lab session** (host `2024ac05133`, Rocky Linux 9.5) the following morning: the disk-full issue had a separate root cause (a broken package build repeatedly re-filling the pipenv cache -- fixed, see `Pipfile`'s `pywinpty` platform marker), but the GPU absence reproduced identically ( `nvidia-smi:`

`command not found`) -- confirming this is a property of the assigned instance, not a transient fault. Screenshots of the blocking condition(s), taken on the Virtual Lab, are attached as proof.

Tasks 2-5 were instead executed locally on an **NVIDIA GeForce RTX 4050 Laptop GPU (6GB VRAM)**, using the same pinned dependency versions (`Pipfile` / `Pipfile.lock`), the same `SEED=42` reproducibility discipline, and the same models (`Qwen/Qwen2.5-1.5B-Instruct`, `Qwen/Qwen2.5-7B-Instruct`) specified throughout this report. All notebook outputs, loss curves, saved adapters, and evaluation numbers in this report are real, executed outputs from that local run -- none were fabricated or estimated.